Lwin MMT

Singapore
+65 93849006 | lwinmmt@gmail.com | linkedin.com/in/lwinmmt | lwinmmt.com

EDUCATION

Singapore Management University Singapore
Bachelor of Science in Information Systems (Product Development & Business Analytics) *August 2023 – Expected 2027*

Singapore Polytechnic Singapore
Diploma in Computer Engineering (networking & cybersecurity) *April 2018 – May 2021*

WORK EXPERIENCE

Vietnam Technology & Telecommunication JSC (VNTT) Ho Chi Minh City, Vietnam
AI & IIoT Engineer Intern *May 2026 – July 2026*

- Standardised simulated Modbus TCP, MQTT and Siemens S7 data for a client's central monitoring needs using an EdgeX Foundry gateway prototype, custom device services and Python simulators in an 18-container Docker stack.
- Implemented deadband filtering to suppress insignificant sensor updates, with enrichment and classification before TLS-secured MQTT export to HiveMQ. Inspected data flow using InfluxDB and Grafana.
- Initiated BecaLearn after identifying scattered intern-onboarding materials, gaining stakeholder support for an LMS centralising documentation and video links.
- Designed and built the LMS for internal hosting, prepared Docker Compose configuration and tested deployment.

Osiris Technology Singapore
Co-Founder *December 2023 – July 2024*

- Built and soldered ESP32 switch-box prototypes with DHT22 sensors and relays for hydroponic lighting and irrigation.
- Implemented ESP32 lighting and irrigation schedules in MicroPython and verified execution without cloud access.
- Presented the BIG grant pitch for Osiris, which secured SMU grant support and incubation.
- Demonstrated the team's remote hydroponic monitoring and control in SME pilots and to visitors at the World Cities Summit 2024 SMU IIE booth.
- Enabled remote monitoring of pilot racks with ESP32 Wi-Fi/MQTT telemetry through HiveMQ Cloud to AWS S3.
- Produced hardware specifications to support the transition from hand-built switch-box prototypes to PCB design.

W2 Industrial Services Hub Singapore
IoT Solutions Consultant (Freelance) *March 2021 – July 2021*

- Enabled remote schedule changes and temperature monitoring for NEA wastewater samplers by integrating SIM-equipped Tastek RTUs, Mosquitto MQTT and an Azure dashboard based on requirements gathered with W2.
- Connected field-device events to WhatsApp equipment alerts for operators by developing a JavaScript backend integrating Mosquitto MQTT with the Twilio WhatsApp API, supporting technician follow-up.
- Set up Inno2's MQTT broker, database and dashboard on one Azure VM for 100+ RTUs across roughly 50 sites.

Singapore Polytechnic Singapore
IoT Engineering Intern *September 2020 – February 2021*

- Led a five-intern IoT team to build a cloud-based dashboard for live monitoring of a 5G autonomous surveillance vehicle using React, Node.js and MySQL.
- Integrated cloud and dashboard components with Raspberry Pi/Node-RED MQTT telemetry for live GPS, battery, temperature, humidity and CO_2 readings.
- Integrated vehicle telemetry into DECADA (Government IoT Cloud Platform) for cloud storage.

SELECTED PROJECTS

Inno2: Wastewater Monitoring & Control: Azure/MQTT system for remote sampler scheduling and monitoring. Contributed to the team's Top 5 recognition in the Lee Hsien Loong IDM Smart Nation Award 2022.

ESMOS: Multi-Cloud Architecture: Designed AWS/Azure recovery architecture to address a course brief's single-provider failure scenario, documenting the added cost and operational complexity in the Service Design Document. Deployed EKS/AKS with the team using Terraform and demonstrated regional traffic failover. Configured BetterStack uptime checks and, with the team, CloudWatch Container Insights and SNS alerts.

TECHNICAL SKILLS

Languages: Python, HTML, CSS, JavaScript/TypeScript, SQL, PowerShell. **Agentic coding:** Codex, Claude Code.
Cloud & deployment: AWS, Azure, Terraform, Docker/Compose, Kubernetes, Linux.
IoT & monitoring: MQTT, EdgeX Foundry, ESP32, Raspberry Pi, Node-RED, InfluxDB, Grafana, MySQL.
Certifications: AWS Certified Solutions Architect - Associate | Microsoft Certified: Azure Fundamentals.